QueryBook: An Illustrated Technical Overview

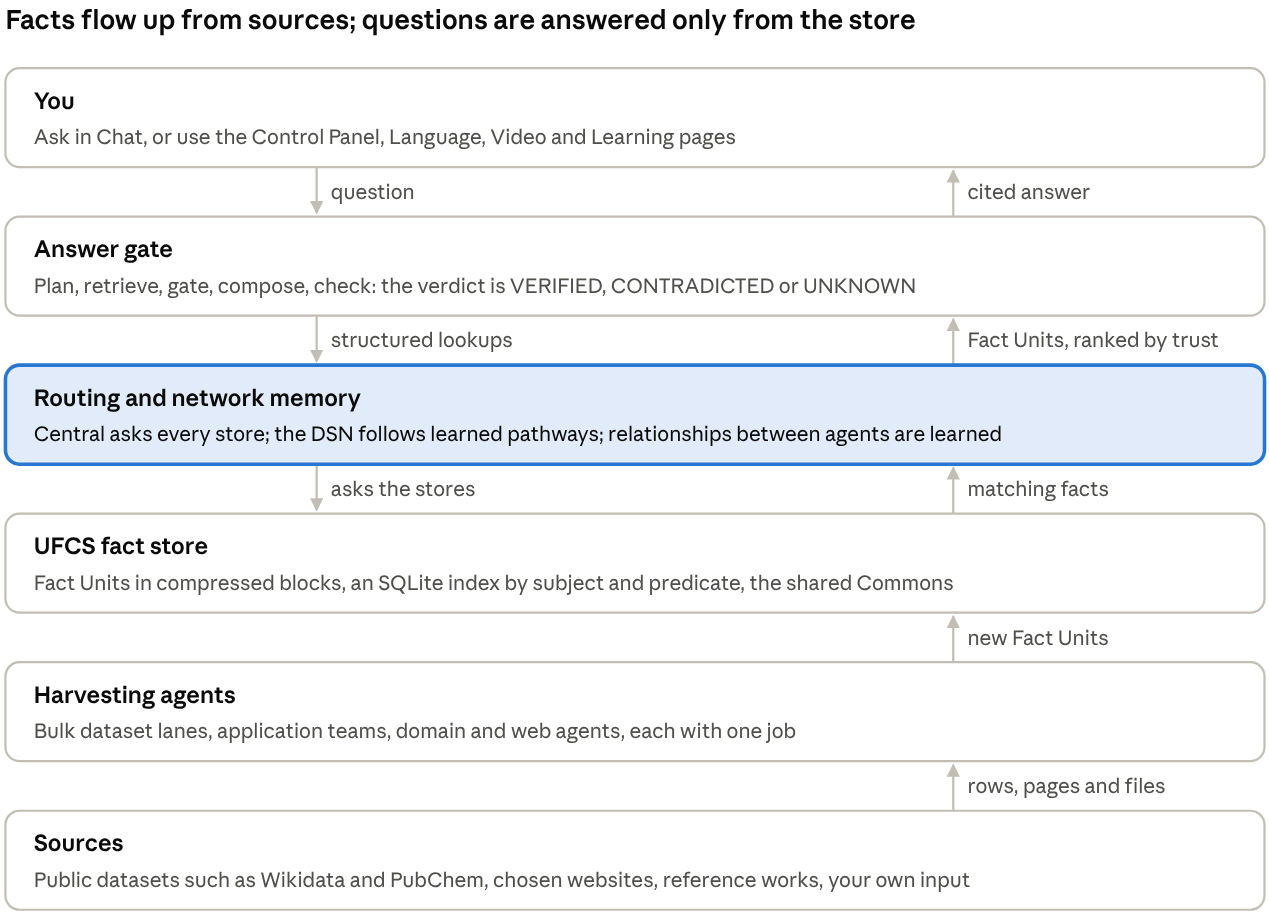
Nov 7, 2026 · @Garry Howard

QueryBook answers questions only from facts it can show and check, and says "I don't know" rather than guess. This overview explains its main ideas in plain words first, then describes each feature.

Part 1: the main ideas

Six ideas explain how QueryBook works: a layered design, Fact Units, the UFCS store, atomic parts, learning connections, and the answer gate.

At a glance

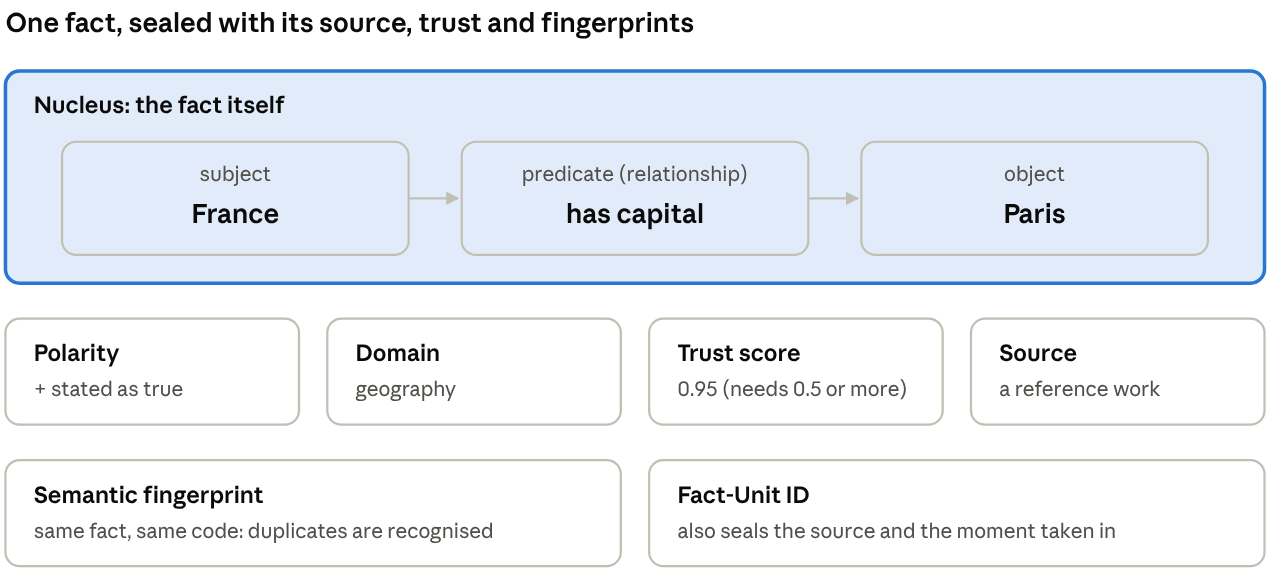


QueryBook at a glance · six layers

Read it from the bottom up for how facts arrive, and from the top down for how a question is answered: nothing reaches an answer without passing through the store and the gate.

Fact Units: the atom of knowledge

A Fact Unit is one fact, stored as one small sealed record: "France — has capital — Paris", plus where it came from and how far it can be trusted. Everything QueryBook knows is made of these, the way a wall is made of bricks.



anatomy of a Fact Unit · nucleus plus six seals

Every Fact Unit has the same parts:

- **The nucleus**: the fact itself, in three pieces — a *subject* (France), a *predicate* or relationship (has capital) and an *object* (Paris).
- **Polarity**: whether the fact is stated as true (+) or as not true (−). "Pluto is not a planet" is a fact too.
- **Domain**: the field it belongs to, such as mathematics, geography, history or language.
- **Sources**: who said it, what kind of authority that is (a reference work, a dataset, a news report) and that source's reputation.
- **Trust score**: a number from 0 to 1. QueryBook only answers from facts above a threshold (0.5 by default); weaker facts are kept but never stated as true.
- **Timestamps and safety**: when it was taken in, and who may read it.
- **Two fingerprints**: a *semantic fingerprint* that is the same for the same fact wherever it comes from, so duplicates are recognised; and a *Fact-Unit ID* that also seals the source and the moment, so this exact record can be proven later.

The fingerprint is a long code computed from the fact's words (after tidying capitals and spaces). Change one letter of the fact and the code changes completely, so a fact cannot be altered quietly. Because the code depends only on the fact, QueryBook can say "we already have this" without comparing millions of records by hand.

UFCS, FQL and SQL: how facts are stored and found

UFCS (the Universal Fact Content Substrate) is the warehouse where Fact Units live; FQL (the Fact Query Language) is how you ask it for them; SQL, through the SQLite database engine, keeps the warehouse's index.

The warehouse (UFCS). Fact Units are packed into large compressed *blocks* on disk, like boxes on shelves. Compression keeps hundreds of millions of facts on an ordinary drive. A *manifest* file keeps the running totals: how many facts, how many duplicates turned away, how many per domain.

The index (SQL). Opening every box to find one fact would be hopeless, so a small SQLite database keeps a card for each fact: its fingerprint, its subject, predicate, object, trust score and which block holds the full record. The cards are sorted by subject and predicate, so a lookup takes milliseconds even across a very large store. SQLite is a standard, file-based database that needs no server to install.

The questions (FQL). FQL asks by structure, not by keywords: "give me facts whose subject is France and predicate is has capital, with trust of at least 0.7". The answer comes back ranked by trust, with a *response provenance hash*: one code summarising exactly which facts were used, so the same question on the same store can be shown to give the same answer. When an exact match finds nothing, QueryBook loosens the search step by step (subject only, then a keyword search) and says what it did.

Many writers at once. Dozens of harvesting agents can add facts at the same moment. A write lock and the database's write-ahead log make each addition complete before the next begins, so no fact is half-written and no counter is lost.

At bulk scale. The suite's bulk harvester keeps whole published datasets (Wikidata, PubChem and others) in their own SQLite *shards*, one per dataset, drawn from the same Fact-Unit idea. A shared database called the *Commons* holds the facts the applications stand behind, with their evidence and reasoning.

Atomic architecture: small, independent, checkable parts

QueryBook is built from atoms at every level: knowledge is split into single facts, and the work is split into small agents that each do one job.

Atomic knowledge. A large language model stores what it learned blended inside billions of numbers, so no one can point to where a statement came from. QueryBook keeps each fact separate, with its own source and trust. That brings four practical gains:

1. **Every answer can cite its facts.** The answer lists the exact Fact Units it used.
2. **One wrong fact can be fixed alone.** Correcting or removing it changes nothing else.
3. **Conflicts are visible.** Two trusted facts that disagree are reported as a contradiction, not quietly averaged.
4. **Duplicates merge.** The same fact arriving from ten sources becomes one fact with ten pieces of evidence.

Atomic work. Harvesting and checking are done by many small agents, each with one task: one per dataset lane, one per domain, a Strategist, Planner, Harvester and Curator in each application team. An agent that fails is restarted without stopping the rest, and more computers can be added by giving them more agents.

Deterministic by design. The core programs are ordinary code, not AI models: the same inputs give the same facts, the same answer and the same fingerprint every time. AI models can be plugged in to help plan a question or phrase an answer, but they never decide what is true — the facts do.

Neuromorphic architecture: connections that learn

QueryBook borrows one lesson from the brain: don't only make the agents smarter, make the connections between them able to learn. It does this in two parts, the Digital Synaptic Network and network memory.

The Digital Synaptic Network (DSN). A brain cell does not shout to every other cell; it signals the few it is connected to, and only when its input crosses a threshold. The DSN routes questions the same way. Instead of asking every store for a fact, it asks the stores its *learned pathways* point to, strongest first. A store that keeps answering for a kind of question gets a stronger pathway; one that keeps coming back empty fades and stops being asked. When the strong pathways find nothing, every store is asked, so routing never hides a fact. Messages between agents are tiny Fact-Unit IDs (12 bytes) rather than paragraphs of text, and each ID is checked against the shared store, which catches an agent that invents a fact.

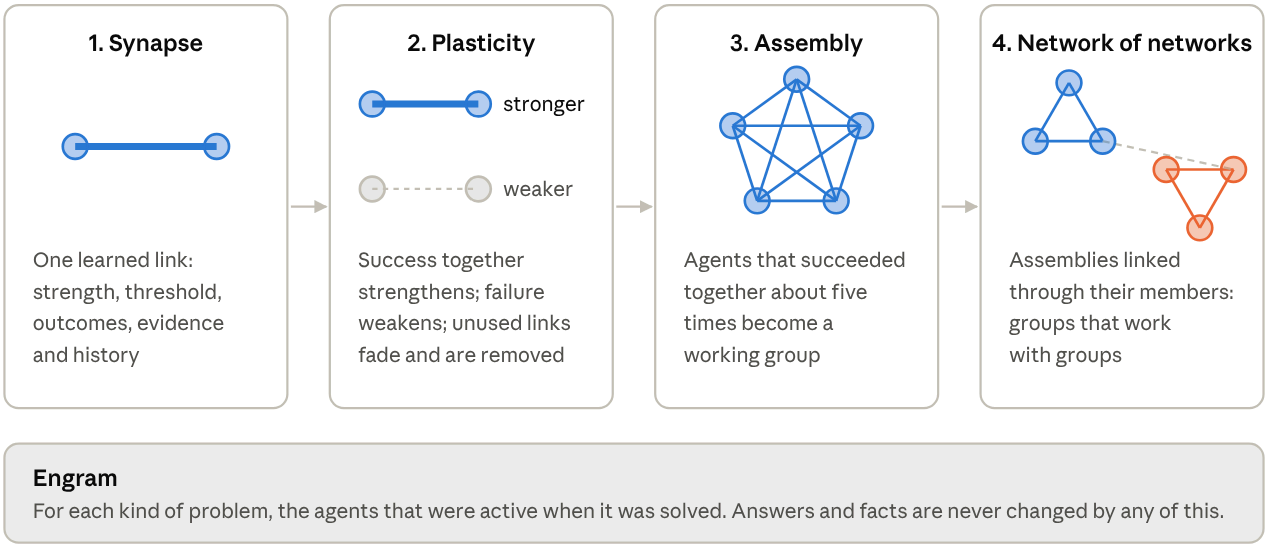
Network memory. The relationships between agents are themselves stored and learned. Each relationship, a *digital synapse*, records:

- its strength, and the threshold it must pass before it carries a signal (higher when it has been unreliable);
- the kinds of problem it formed in, its good and bad outcomes, the latest evidence and a history of changes.

It learns the way synapses do. Agents that succeed together strengthen their link; a joint failure weakens it; an unused link fades and is eventually removed; a link with a long good record is *consolidated* and fades far more slowly. One coincidence is not a relationship: a new link starts weak.

From thousands of such links, larger structures appear. Agents that have solved problems together about five times form an *assembly*, a working group. Assemblies linked through their members form *networks of networks*. For each kind of problem, QueryBook keeps an *engram*: the agents that were active when it was solved.

Single learned links add up to working groups and networks of networks



network memory · from one synapse to networks of networks

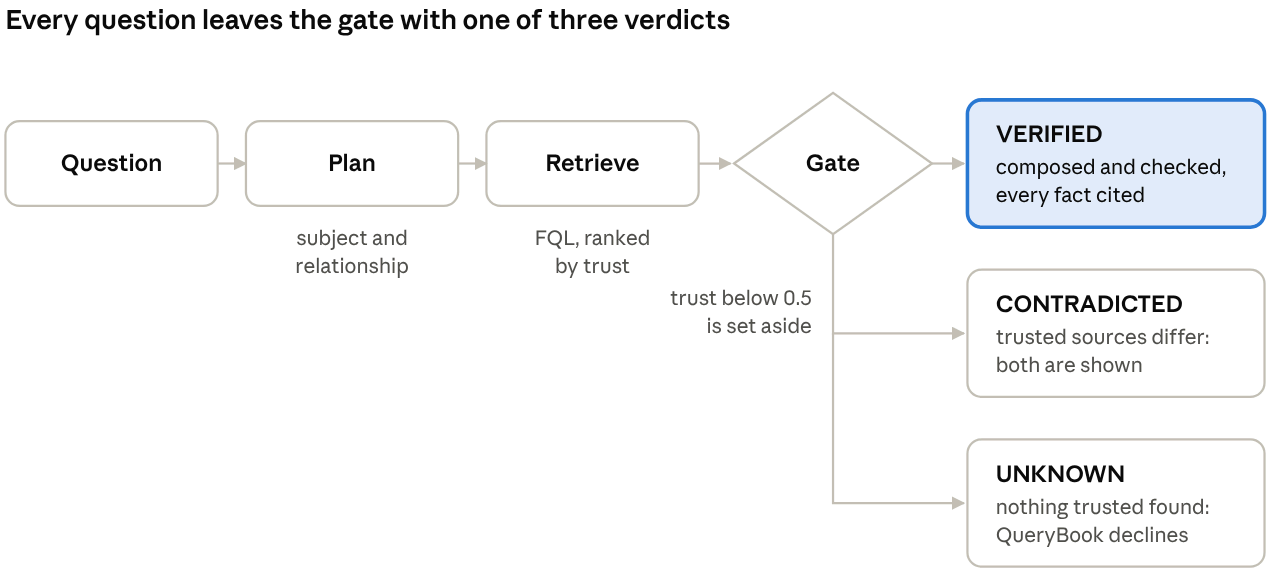
What it does and does not do. Network memory records, adapts and explains how agents work together, so routing and review can use it. It never changes an answer or a fact. It does not create intelligence by itself; neuroscience has not shown that such a hierarchy alone produces intelligence either.

What the stress test found. A simulation compared six engines answering the same questions under extreme stress (lying, dead and slow agents; up to 10,560 agents; a million facts). The DSN beat conventional multi-agent messaging by a wide margin, using about 5 messages per question instead of hundreds or thousands, and it gave no wrong answers where broadcasting gave many. On one computer, though, the central pipeline was still faster and slightly more accurate. The DSN pays off when facts are spread over many stores or computers, and its learning helps most once agents also learn new verified facts.

Using it safely. In the Control Panel, Answer routing has three settings: *Central* (ask every store, the default), *Shadow* (Central answers while the DSN runs beside it and every difference is logged) and *DSN*. The health report gives a verdict from the Shadow log before anyone switches.

Trust and the answer gate: cite or decline

Every answer passes through a gate with only three exits: VERIFIED (answered, with citations), CONTRADICTED (trusted sources disagree, and both are shown) or UNKNOWN (QueryBook declines rather than guess).



the answer gate · three steps in, three verdicts out

A question goes through five steps:

1. **Plan.** The question is turned into one or more structured lookups (a subject and a relationship). An AI model can help here when one is configured; without one, simple rules do it.
2. **Retrieve.** FQL fetches the matching Fact Units, ranked by trust, and records the response provenance hash.
3. **Gate.** Facts below the trust threshold are set aside. If nothing trusted remains, the verdict is UNKNOWN. If trusted facts give different values for something that can only have one value (a capital, a birth date), the verdict is CONTRADICTED. Relationships that can rightly have several values ("is a", "part of") are not treated as conflicts.

4. **Compose.** The answer is written from the gated facts only, each one numbered so the answer can cite it.
5. **Check.** Any sentence in the draft that no cited fact supports is flagged, so an AI model's wording cannot slip in a claim of its own.

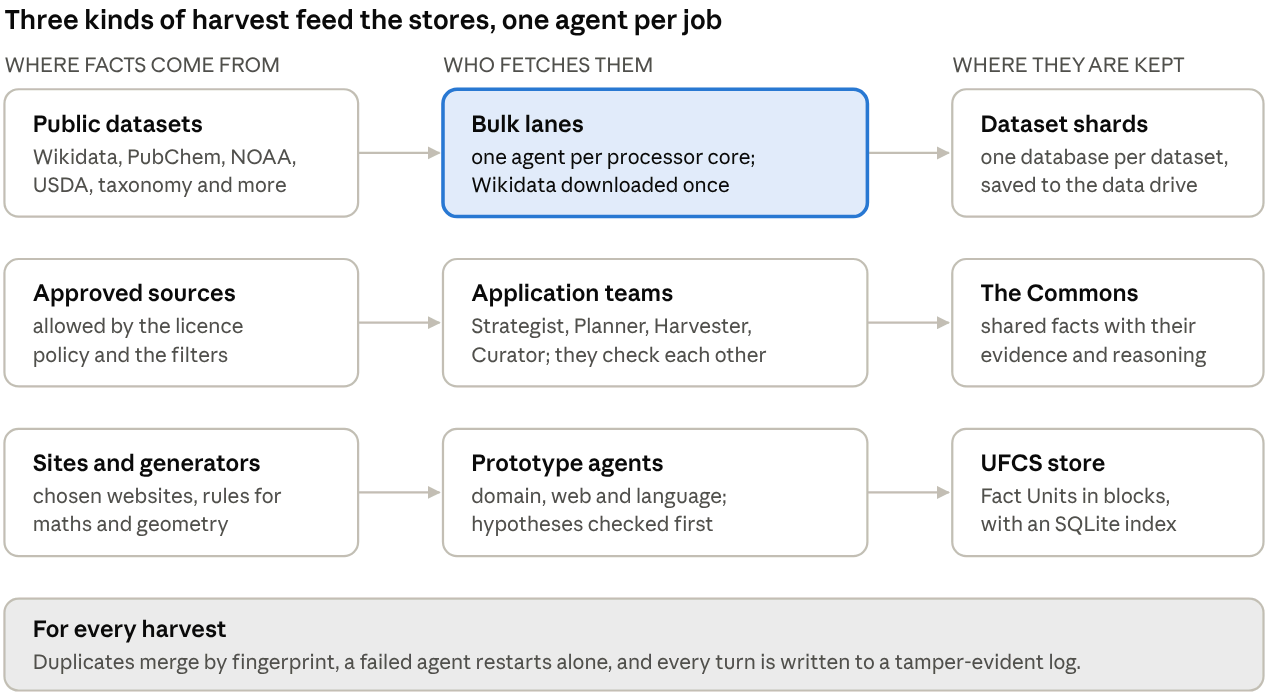
This is the *Prime Directive*: QueryBook may only state what it can show. A wrong answer is the worst outcome; "I don't know" is safe.

Part 2: the features, one by one

QueryBook comes in two forms built on the same ideas: the **prototype** (QueryBook 9.64, one program with many pages) and the **suite** (five programs that share one fact database). The features below are grouped by what they do: gather facts, answer, teach and speak, keep things safe, and run the system.

Harvesting: how facts come in

Harvesting is done by agents that run around the clock, each turning one source into Fact Units; three kinds of harvest cover different sources.



harvesting · three kinds, from source to store

Bulk harvesting (the fast lane). Whole public datasets are read from start to finish, each row becoming facts:

Dataset	What it gives
Wikidata (16 lanes)	The structured facts behind Wikipedia: people, places, works, organisations
PubChem	Chemical compounds and their properties
NCBI Taxonomy	The names and family tree of living things
NOAA weather	Daily weather records from stations worldwide
USGS place names	Official geographic names
USDA FoodData	Foods and their nutrients
Tatoeba	Sentences and their translations
MIT OpenCourseWare	Course material listings
Calendar	Computed date facts

One agent runs per processor core. Each writes first to a fast working copy on the computer's internal drive, which is saved to the data drive every few minutes. A *Find and restore missing facts* step compares every copy and adds back anything the live copy lacks; nothing is ever deleted.

Download once. Wikidata's full file is about 157 GB. Rather than 16 lanes each fetching it over the internet, it is downloaded once, from several servers at the same time, checked against the publisher's checksum, and then read from the drive. While the download runs, the lanes stream over the internet so harvesting never stops, sharing each server's connection limit with the download.

Application teams. Each suite application runs a team of four agents: a *Strategist* (what is missing and which sources are allowed), a *Planner* (concrete tasks), a *Harvester* (fetches and stages facts) and a *Curator* (saves them, merges duplicates, answers checks other applications asked for, and takes backups). Applications cooperate only through the shared database: one can ask another to check a fact ("is this lifespan possible?"), and the answer becomes evidence on the fact.

Prototype agents and campaigns. The prototype runs its own agents: domain generators (mathematics, geometry), web harvesters for chosen sites, language agents, and *hypothesis* agents that may only propose facts, which are then checked against the store before being kept apart. Harvest campaigns draw on 23 open online sources under one master on/off switch.

Rules on what may enter. A licence policy decides which sources may be stored; filters can leave out facts that mention certain words; every harvest turn is written to a tamper-evident log.

The QueryBook suite: five programs, one fact database

The suite is five programs that open in a web browser on your own computer; four share one fact database, the Commons, and none needs an internet account to run.

Program	Address	What it does
Control Panel	127.0.0.1:8100	One page for everything: each program's health, the harvest switch, bulk harvesting, answer routing, targeting and filters, shared facts, the audit trail, diagnostics, connections and keys, and the health report
Language Builder	127.0.0.1:8101	Words and speech: vocabulary cards, translation, pronunciation, voices, talking clips made by Gemini (HeyGen as fallback), and lip reading
Math, Science & Engineering	127.0.0.1:8102	Exact calculation with units, formulas solved for any unknown, physical constants, the periodic table and molar masses; saved answers are re-checked by recomputing them
History & News	127.0.0.1:8103	Dated facts about events and people, with their sources and feeds
Video Studio	127.0.0.1:8104	Narrated videos built from facts, with voices and narrators from Language Builder

The Commons. Each program keeps its own working data, but the facts it is prepared to stand behind go into the Commons, where the other programs can read them, check them and add evidence. Every write is one complete transaction, the database checks itself at start-up, and verified backups are taken.

Health and transparency. The Control Panel raises an alert when a program stops unexpectedly or a harvest fails. Every action goes into an audit trail. A plain-language health report (also sent by email if you choose) says how fast facts are arriving, the main delay and what is being done about it, what recovered by itself, the answer routing and network memory, and what, if anything, you need to do.

Your data stays yours. Facts and settings live outside the program folders (on a data drive such as D:\QB), so installing a new version never touches them. Voice cloning works only behind an explicit consent step.

The prototype's features

The prototype (QueryBook 9.64) is one program with a page for each area; its feature catalogue lists 42 features, all checked live by its own self-test.

Bringing facts in

Feature	Page	What it does
Ingestion ladder	/ingest	Runs ingestion in automated stages, from simple sources to harder ones
Domain generators	/ingest	Mathematics, geometry and arithmetic facts, generated and growing each cycle
History, news, culture and art domains	/ingest	Seeded domain facts, stored as attributed claims ("recorded by" a named source)
Web harvester	/ingest	Crawls chosen sites politely and extracts Fact Units
Language ingestion	/language	Builds vocabulary, definitions and pronunciations in stages
Open Claw ingestion	/openclaw	Takes in arbitrary files and splits them into checkable pieces

Feature	Page	What it does
Direct ingestion	—	Adds Fact Units from questions and answers or by hand

Asking

Feature	Page	What it does
Cited chat	/chat	Answers with citations, or UNKNOWN; now also reports the specialists' assembly
FQL and Fact-Unit lookup	/chat	Structured queries and the full record of any fact
Query understanding	—	Finds the entities, relationship and intent in a question
Network memory	/network	The learned relationships between agents: assemblies, engrams, strongest links

Language, voice and video

Feature	Page	What it does
Translate, detect, pronounce, speak	/language	Language tools grounded in stored facts
Dialects	/language	Six dialect profiles
Cloud voices and cloning	/language	Google and ElevenLabs voices; cloning only with consent
Voice Lab	/voice	Two-way voice, tone and voice matching
Voiceprint (LIL)	/voice	Enrol, match and authenticate a voice, with consent
Paralinguistics (PIL)	/voice	Reads tone and intent beyond the words
Scene Director	/director	Turns QueryBook content into direction for Gemini video
Scene Reconstructor	/reconstructor	Turns prose into a directed video clip

Learning

Feature	Page	What it does
Learner experience	/learn	Lessons, quizzes and mastery tracking
Educator analytics	/educator	Class and cohort mastery at a glance
Collaboration	/collab	Shared annotations, threads and resources
Ontology and assessment	—	A concept graph, assessments and study guides

Integrity and security

Feature	Page	What it does
Rights gate	—	Checks licensing before content is stored or shown
Provenance ledger	—	Seals content and proves where it came from
Provenance firewall	—	Middleware that refuses unprovenanced content
NFT engine	/openclaw	Optional on-chain records of facts, answers and their lineage
Agent protection	/security	Gateway, decoys, tripwires and anomaly detection against hostile agents
Immutable security log	/security	A hash-chained log with checkpoints that cannot be rewritten

Running the platform

Feature	Page	What it does
Two-system home	/	Ingestion on one side, Query on the other
Editions	/	Light, medium and full
Embodiments	/	Chatbot, engine, command line, library, MCP server, middleware, SDK
Reasoning agents and campaigns	/dashboard	Agents, harvest campaigns and the master on/off switch

Feature	Page	What it does
Self-knowledge	/self	QueryBook describes itself accurately, with sources
Multi-tenant admin	/admin	Tenants, users, roles, usage and plans
Open Claw agent support	—	Works with agent tools through an MCP gateway

Glossary

Term	Meaning
Agent	A small program with one job, such as harvesting one dataset or checking one kind of fact
Assembly	A group of agents bound by strong links after succeeding together repeatedly
Commons	The suite's shared fact database, with evidence and reasoning for each fact
Consolidation	A link with a long good record that fades much more slowly
CONTRADICTED	The verdict when trusted sources disagree; both are shown
Digital synapse	A learned relationship between two agents, with strength, threshold, outcomes, evidence and history
Download once	Fetching a large dataset file one time and reading it from the drive
DSN	Digital Synaptic Network: routing that asks only the stores its learned pathways point to
Engram	For one kind of problem, the agents that were active when it was solved
Fact Unit	One fact as a sealed record: subject, predicate, object, sources, trust, fingerprints

Term	Meaning
Fingerprint	A code computed from a fact's words; the same fact always gives the same code
FQL	Fact Query Language: asking for facts by subject, relationship and minimum trust
Lane	One of the parallel agents reading a share of a large dataset
Network memory	The stored, learning relationships between agents
Prime Directive	QueryBook may only state what it can show; otherwise it declines
Provenance hash	One code summarising exactly which facts an answer used
Shadow	A routing setting where Central answers and the DSN is compared alongside
Shard	One dataset's own database file in the bulk harvest
SQL / SQLite	The standard database language and the file-based engine that keeps QueryBook's index
Trust score	A number from 0 to 1; only facts above the threshold (0.5 by default) are stated
UFCS	Universal Fact Content Substrate: the compressed store where Fact Units live
UNKNOWN	The verdict when no trusted fact answers the question
VERIFIED	The verdict when trusted facts answer the question; the answer cites them